

Juan Antonio G. Soria Rodríguez

Software Developer & AI Researcher | NLP and LLMs

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Professional Summary

Final-year Computer Science student at Universidad Autónoma Metropolitana (UAM-I, degree expected December 2026), specialized in Artificial Intelligence, Natural Language Processing (NLP), and AI-augmented software development. Completed a research-grade Bachelor's thesis building a production-level, dual-mode paraphrase and plagiarism detection system over an institutional corpus of ~5,000 theses — combining deep learning, vector retrieval, transformer fine-tuning, and containerized web deployment.

Actively uses AI-augmented methodologies: context engineering, agentic workflows, AI-assisted code generation, and multi-agent system design. Views AI as a strategic multiplier for analyzing complex problems, validating solutions, and transforming ideas into functional products.

Education

B.Sc. in Computer Science

2021 – 2026

[Universidad Autónoma Metropolitana \(UAM-I\), Mexico City](#)

Thesis (Capstone Projects I & II — degree expected Nov–Dec 2026): Intelligent Paraphrase Detection using Large Language Models (LLMs)

Advisor: Dr. Benjamín Moreno Montiel

Capstone II (in progress): sentiment analysis of the Codex Gigas — NLP for Latin with contextual language models

Applied Mathematics and Computing

2011 – 2016

[Universidad Nacional Autónoma de México \(UNAM\), FES Acatlán](#)

Capstone Project I: Paraphrase & Plagiarism Detection System

End-to-end, dual-mode NLP system (pairwise classification + corpus search) for academic plagiarism detection at UAM-I. Deployed over ~5,000 theses with a FAISS semantic index of **2.28M vectors at 100% corpus coverage**.

- **Architecture:** Bi-encoder MiniLM + FAISS → cross-encoder XLM-RoBERTa/BETO → optional LLM arbiter (Llama 3.3 70B) in the decision boundary [0.35, 0.65].
- **Models:** Fine-tuned XLM-RoBERTa (calibrated, T=5.15, ECE 0.076), BETO v8a (F1=0.889), soft-voting ensemble with **AUROC 0.999**. Rigorous evaluation with bootstrapped confidence intervals and data-leakage audit.
- **Key finding:** The CBI corpus is ~97% original (only ~3% real paraphrase). Core contribution is data efficiency: **60 human pairs match 180K algorithmic pairs**.
- **Delivery:** Unified web app with Docker, PWA, section-level similarity reports, heatmaps, and LLM-powered explainability.

Stack: Python, PyTorch, Hugging Face Transformers, sentence-transformers, FAISS, MongoDB, Docker.

Technical Skills

Languages:	Python (Advanced)		Java (Experienced)	C / C++	JavaScript	PHP	
ML / NLP:	PyTorch	Hugging Face	BETO / XLM-R	sentence-transformers	FAISS	LLMs / RAG scikit-learn	
AI-Augmented:	Prompt Engineering		Context Engineering	Agentic AI	AI-Assisted Code	Multi-Agent	
Tools:	Git / GitHub		Docker	MongoDB	MySQL	Linux	Jupyter / Colab
Web / Backend:	Spring Boot	Next.js / React	FastAPI	Node.js	PostgreSQL	REST / Microservices	

Professional Experience

Freelance Software Developer Independent	2022 – Present
Custom web systems for clinics and small businesses — management, scheduling and invoicing — built with Spring Boot, Next.js and PostgreSQL, deployed to production.	
Adjunct Instructor (freelance) Admission and remedial courses	Feb 2020 – Dec 2021
High-school and university admission courses plus Algebra, Geometry, Calculus and Physics; designed course content and adapted it to each group.	
IT Support Technician BTL Company	Apr 2017 – Apr 2021
Computer maintenance, software and OS installation, and day-to-day user support.	
Security Systems Installer SYSARK	Jul 2016 – Aug 2019
Installation and configuration of CCTV and access-control systems; on-site project coordination and delivery.	

Highlighted Personal Projects

Full-Stack Management Systems
End-to-end systems for veterinary and dental clinics: Spring Boot 3 + PostgreSQL + JWT + WebSockets (interactive odontogram, PDF invoicing) and a Next.js 15 + Turborepo + Drizzle ORM monorepo deployed on Vercel.
Microservices Architecture
E-commerce and pharmacy platforms with up to 8 Spring Boot microservices , RabbitMQ, API Gateway, and Docker Compose orchestration.
Artificial Intelligence & Algorithms (Java)
Portfolio of 15+ projects: BFS, DFS, A*, Minimax, Negamax, and Alpha-Beta pruning. Game logic for Connect 4, Tic-Tac-Toe, N-Queens, and sliding puzzles using Java Swing.
Low-Level Systems & Compilers (C)
Spanish→C pseudocode compiler with Flex & Bison (lexical, syntactic, semantic analysis, and code generation). Operating Systems and Concurrent Programming projects in C (processes, threads, shared memory).
Game Development
2D Mario-style platformer in Java (Swing, MVC architecture, Gradle) and a low-poly 3D open world in Unity / C# .

Relevant Coursework

Linear Algebra Calculus Discrete Mathematics Probability & Statistics Software Engineering Operating Systems AI Machine Learning Databases Discrete Simulation Networks Algorithm Design & Analysis
Languages: Spanish (Native) English (Intermediate / Technical reading & writing) Portuguese (Basic)
Mexican nationality. Available for junior roles, internships, freelance, and research initiatives in ML/AI engineering and NLP.